

DDI-Guard — Explainable AI for Drug–Drug Interaction Risk Assessment in Pharmaceutical R&D

Team Num	:	T012
Team Name	:	Wizards
Problem Theme	:	Explainable AI for Drug–Drug Interaction (DDI) risk prioritization in pharmaceutical R&D
Institute Name	:	Thapar Institute of engineering and technology
Team Members	:	Mannat, ronit, jaspreet, jiya, arshpreet, mayuri
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1. Background

- *Pharmaceutical companies must assess drug–drug interaction (DDI) liabilities throughout discovery and development — long before a compound has real-world clinical exposure.*
- **Combinatorial Explosion**: Just **2,310** approved drugs create **~2.7 million** possible interaction pairs.
- **New or poorly** characterized candidate compounds typically have little to no historical interaction evidence, yet the combinatorial space of possible drug pairs is enormous.
- **The Database Gap**: Resources like DrugBank and TWOSIDES only document known, previously observed interactions.
- **End User**: Pharmaceutical R&D, clinical pharmacology, drug safety/pharmacovigilance, translational medicine, and cheminformatics teams — not consumers or prescribers.

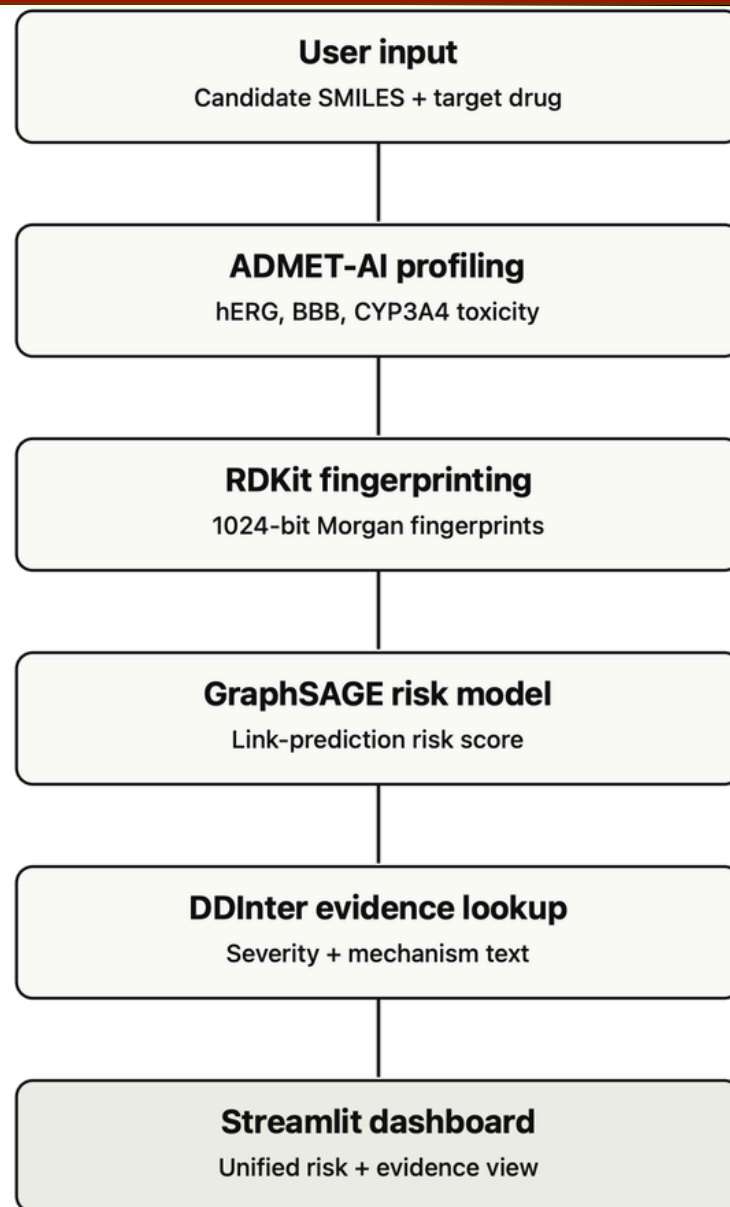
2. Problem Statement

How might we help pharmaceutical researchers prioritize drug pairs for further DDI investigation using explainable, evidence-based risk predictions?

- *Unmanageable Scale*: Exhaustive experimental lab validation is too expensive and impossible to run on every pair.
- *Manual Bottlenecks*: Cross-referencing chemical structures and literature manually simply does not scale.
- *The Trust Deficit*: Traditional "black-box" machine learning is unhelpful to researchers if they cannot see why a pair was flagged.
- *Our Mission*: We need an explainable, evidence-grounded AI system to prioritize which combinations require lab validation first.

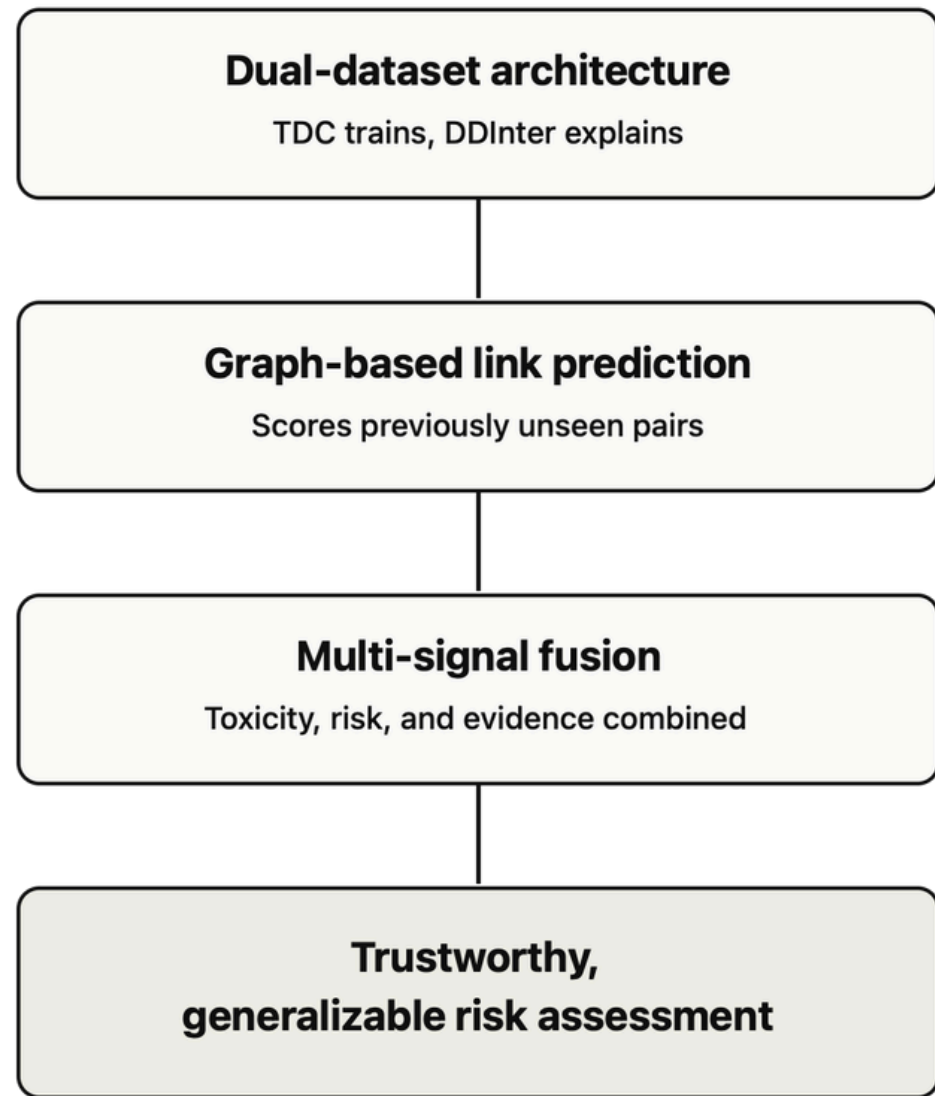
3. Proposed Solution

- **Input** — user enters a candidate drug (SMILES) and selects a target drug from the database
- **Toxicity profiling** — ADMET-AI predicts cardiac (hERG), blood-brain barrier, and CYP3A4 metabolic risk for the candidate
- **Molecular fingerprinting** — RDKit converts both drugs into 1024-bit Morgan fingerprints, a machine-readable structural signature
- **Risk prediction** — a GraphSAGE graph neural network scores interaction likelihood via link prediction on a drug-interaction graph
- **Clinical evidence** — DDInter is queried for documented interaction severity and mechanism, when available
- **Output** — all three signals are unified in a single Streamlit dashboard for interpretable, actionable risk assessment



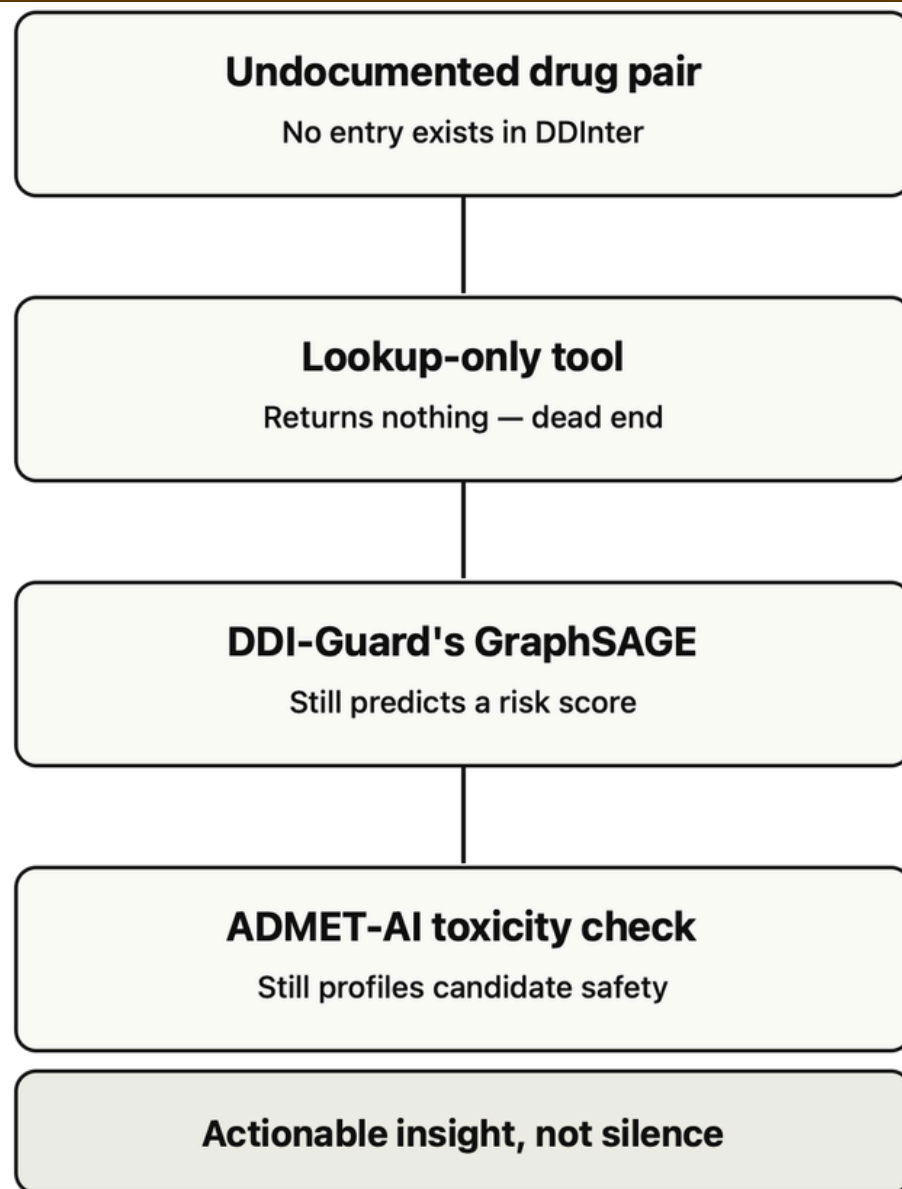
4. Novelty

- **Dual-dataset architecture** — TDC DrugBank DDI trains the model (clean, paired SMILES)
- **DDInter is used only for evidence display** — kept fully out of the training loop to prevent data leakage
- **Graph-based link prediction** — unlike a static lookup table, GraphSAGE generalizes to drug pairs never seen during training, based on learned structural/chemical similarity
- **Multi-signal fusion** — toxicity profile, predicted interaction risk, and real clinical evidence are shown together, not as three disconnected outputs
- **Result** — a system that's both trustworthy (grounded in real clinical data where it exists) and generalizable (still useful where it doesn't)



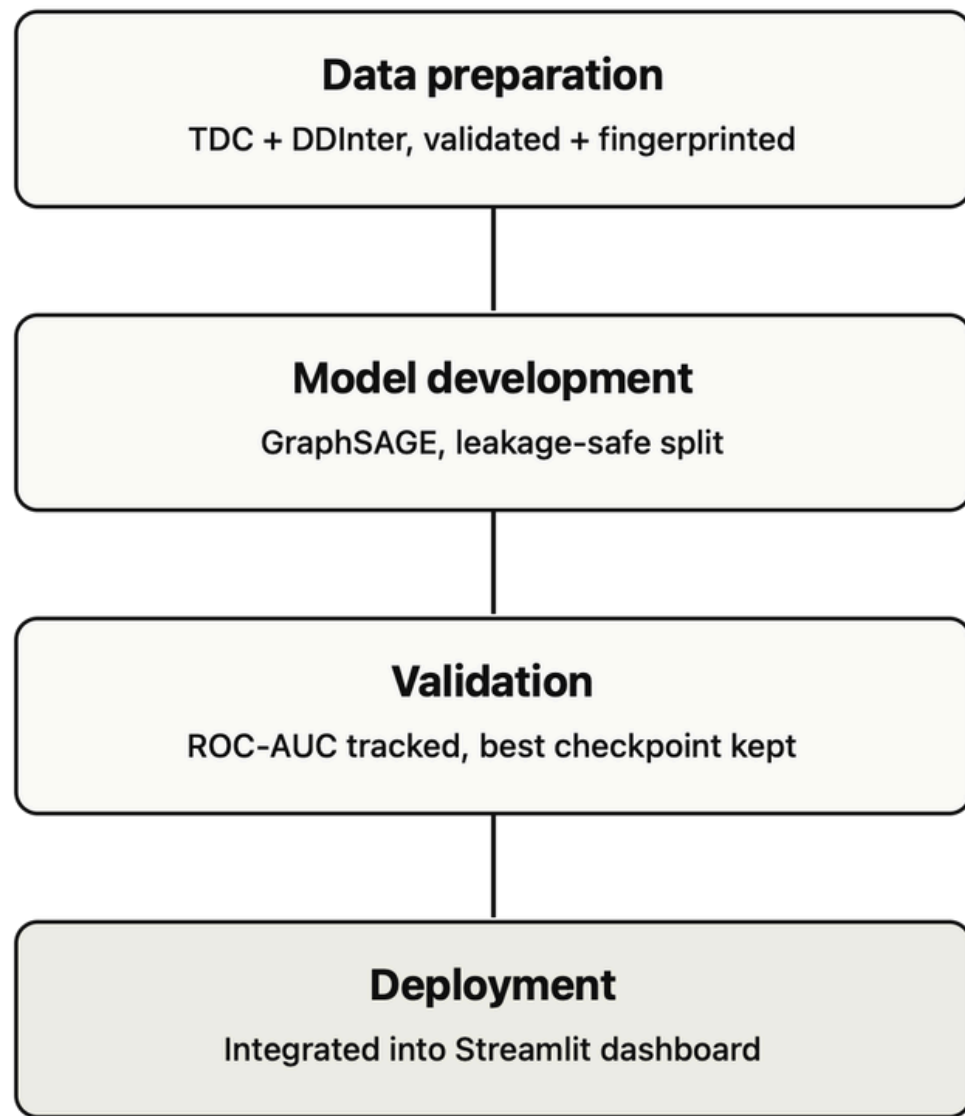
5. Show Stopper

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6. Methodology

- **Data preparation** — sourced TDC DrugBank DDI (training) and DDInter (evidence) as separate datasets, validated SMILES, generated Morgan fingerprints
- **Model development** — built the drug-interaction graph and trained a GraphSAGE link-prediction model with a leakage-safe train/test split
- **Validation** — tracked ROC-AUC across training epochs, kept the best-performing checkpoint
- **Deployment** — integrated the trained model with ADMET-AI toxicity scoring and DDInter evidence lookup into one Streamlit dashboard



7. Team Role

Role → Team ↓	ML	Backend	Database	Chem- Informatics	frontend
Mannat	✓	✓			
Jaspreet	✓	✓			✓
Ronit	✓	✓	✓		
Jiya				✓	
Arshpreet				✓	
Mayuri				✓	

8. References

- Landrum, G. et al. RDKit: Open-source cheminformatics. <https://www.rdkit.org>
- Swanson, K., Walther, P., Leitz, J., Mukherjee, S., Wu, J. C., Shivnaraine, R. V., & Zou, J. (2024). ADMET-AI: a machine learning ADMET platform for evaluation of large-scale chemical libraries. *Bioinformatics*, 40(7), btae416. <https://doi.org/10.1093/bioinformatics/btae416>
- Hamilton, W. L., Ying, R., & Leskovec, J. (2017). Inductive Representation Learning on Large Graphs. *Advances in Neural Information Processing Systems (NeurIPS)* 30, 1024–1034.
- Veličković, P., Cucurull, G., Casanova, A., Romero, A., Liò, P., & Bengio, Y. (2018). Graph Attention Networks. *International Conference on Learning Representations (ICLR)*.
- Xiong, G., Yang, Z., Yi, J., Wang, N., Wang, L., Zhu, H., Wu, C., Lu, A., Chen, X., Liu, S., Hou, T., & Cao, D. (2022). DDInter: an online drug–drug interaction database towards improving clinical decision-making and patient safety. *Nucleic Acids Research*, 50(D1), D1200–D1207. <https://doi.org/10.1093/nar/gkab880>
- Streamlit Inc. Streamlit: A faster way to build and share data apps. <https://streamlit.io>
- Huang, K. et al. (2021). Therapeutics Data Commons: Machine Learning Datasets and Tasks for Drug Discovery and Development. *NeurIPS Datasets and Benchmarks Track*. <https://tdcommons.ai>



Thank You